

Problem overview:

Given the population of N seabirds nesting in a sanctuary composed of island East (E) and island West (W), we have the following proportions:

$p_E \Rightarrow$ proportion of seabirds nesting on island E in year 1

$p_W = 1 - p_E \Rightarrow$ proportion of seabirds nesting on island W in year 1

Considering that some proportion of seabirds may migrate to a different island, we have:

$\delta \Rightarrow$ probability that a seabird will change from island E to W and vice versa

$\delta = P(\text{the seabird is on island E in year 2} \mid \text{the seabird is on island W in year 1})$

$= P(\text{the seabird is on island W in year 2} \mid \text{the seabird is on island E in year 1})$

Since p_E and δ are unknown, we assume that the two quantities have the following priors that are independent of each other:

$$p_E \sim U(0, 1) \Rightarrow \frac{1}{1-0} = 1$$

$$\delta \sim \text{Beta}(2, 10) \Rightarrow \frac{\Gamma(12)}{\Gamma(2)\Gamma(10)} \delta(1-\delta)^9$$

During year 2, it was observed that a total of $XE = xE$ out of the N seabirds were on island E and the remaining $N - xE$ seabirds were on island W.

1. Evaluating the probability that a seabird is on island E in year 2 (θ_E), and the probability that a seabird is on island W in year 2 (θ_W):

$$\theta_E = p_E(1 - \delta) + p_W\delta$$

$$\theta_W = p_W(1 - \delta) + p_E\delta$$

Explanation:

θ_E :

$p_E(1 - \delta)$ calculates the probability of a seabird remaining on island E for both year 1 and 2

$p_W\delta$ calculates the probability of a seabird transitioning from island W for year 2

θ_W :

$p_W(1 - \delta)$ calculates the probability of a seabird remaining on island W for both year 1 and 2

$p_E\delta$ calculates the probability of a seabird transitioning from island E for year 2

Therefore we can also assume that $\theta_W = 1 - \theta_E$ since we are only given 2 islands, hence if a bird is not on island E, it can be assumed that it is on island W and vice versa.

2. Evaluating the following probabilities:

τ_E : the probability that a seabird is on island E in year 1 given that it is on island E in year 2

τ_W : the probability that a seabird is on island W in year 1 given that it is on island W in year 2

To estimate τ_E , let,

A = the probability that the seabird is on island E in year 1

B = the probability that the seabird is on island E in year 2

Hence by applying the Bayes theorem, we have:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \therefore \tau_E = \frac{(1-\delta)p_E}{\theta_E}$$

To estimate τ_W , let,

A = the probability that the seabird is on island W in year 1

B = the probability that the seabird is on island W in year 2

Hence by applying the Bayes theorem, we have:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \therefore \tau_W = \frac{(1-\delta)p_W}{\theta_W}$$

3. Assessing the marginal distribution of X_E

Since X_E denotes the number of seabirds on island E in year 2, we can conclude that every seabird that ends up on island E in year 2 is a success, therefore we can assume that X_E follows a Binomial distribution.

$$X_E \sim \text{Bin}(N, \theta_E)$$

$$\Rightarrow p_{X_E}(X_E) = \binom{N}{X_E} \theta_E^{X_E} (1 - \theta_E)^{N-X_E} \quad \text{where } 0 \leq X_E \leq N \text{ and } 0 < \theta_E < 1$$

In which case, N is the total number of birds and θ_E is the probability of transitioning to island E in year 2.

4. Given that Y_E = the number of seabirds on island E for year 1 and 2, we can assume that the distribution of Y_E conditional to (X_E, p_E, δ) is Binomial with the following parameters:

$$Y_E | (X_E, p_E, \delta) \sim \text{Bin}(X_E, \tau_E)$$

The probability τ_E is applicable since we are looking for the number of seabirds that were on island E for both years. Therefore, we obtain the following PMF:

$$p_{Y_E}(Y_E | X_E, p_E, \delta) = \binom{X_E}{Y_E} \tau_E^{Y_E} (1 - \tau_E)^{X_E - Y_E} \quad \text{where } 0 \leq Y_E \leq X_E \text{ and } 0 < \tau_E < 1$$

Given that Y_W = the number of seabirds on island W for year 1 and 2, we can assume that the distribution of Y_W conditional to (X_E, p_E, δ) is Binomial with the following parameters:

$$Y_W | (X_E, p_E, \delta) \sim \text{Bin}(N - X_E, \tau_W)$$

The probability τ_W is applicable since we are looking for the number of seabirds that were on island W for both years. Therefore, we obtain the following PMF:

$$p_{Y_W}(Y_W | X_E, p_E, \delta) = \binom{N - X_E}{Y_W} \tau_W^{Y_W} (1 - \tau_W)^{N - X_E - Y_W} \quad \text{where } 0 \leq Y_W \leq N - X_E \text{ and } 0 < \tau_W < 1$$

5. Given that Y_E and Y_W are independent of each other and are dependent on X_E , we can evaluate a full conditional distribution $\pi(p_E, \delta | x_E, y_E, y_W)$ using X_E, Y_E, Y_W .

Defining likelihood functions of X_E, Y_E, Y_W :

$$\begin{aligned} L(X_E) &= \binom{N}{x_E} \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \\ L(Y_E) &= \binom{x_E}{y_E} \tau_E^{y_E} (1 - \tau_E)^{x_E-y_E} \\ L(Y_W) &= \binom{N-x_E}{y_W} \tau_W^{y_W} (1 - \tau_W)^{(N-x_E)-y_W} \end{aligned}$$

Defining priors for p_E and δ :

$$\begin{aligned} p_E &\sim U(0, 1) \Rightarrow \frac{1}{1-0} = 1 \\ \delta &\sim \text{Beta}(2, 10) \Rightarrow \frac{\Gamma(12)}{\Gamma(2)\Gamma(10)} \delta(1 - \delta)^9 \end{aligned}$$

Knowing the information above, we can construct the conditional distribution through the conditional probability:

$$\begin{aligned} f(Y_E, Y_W | X_E) &= \frac{f(Y_E, Y_W, X_E)}{f(X_E)} \\ \therefore f(Y_E, Y_W, X_E) &= f(Y_E, Y_W | X_E) f(X_E) = f(Y_E | X_E) f(Y_W | X_E) f(X_E) \\ &= \binom{x_E}{y_E} \tau_E^{y_E} (1 - \tau_E)^{x_E-y_E} \cdot \binom{N-x_E}{y_W} \tau_W^{y_W} (1 - \tau_W)^{(N-x_E)-y_W} \cdot \binom{N}{x_E} \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \end{aligned}$$

Where, $L(Y_E, Y_W, X_E) = f(Y_E, Y_W, X_E)$

Therefore using the posterior formula we obtain the following full conditional distribution:

$$\pi(p_E, \delta | x_E, y_E, y_W) \propto L(Y_E, Y_W, X_E) \cdot \pi(\delta) \cdot \pi(p_E)$$

$$\begin{aligned} \pi(p_E, \delta | x_E, y_E, y_W) &= \binom{x_E}{y_E} \tau_E^{y_E} (1 - \tau_E)^{x_E-y_E} \cdot \binom{N-x_E}{y_W} \tau_W^{y_W} (1 - \tau_W)^{(N-x_E)-y_W} \\ &\quad \cdot \binom{N}{x_E} \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \cdot \frac{\Gamma(12)}{\Gamma(2)\Gamma(10)} \delta(1 - \delta)^9 \cdot 1 \\ &\propto \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \cdot \tau_E^{y_E} (1 - \tau_E)^{x_E-y_E} \cdot \tau_W^{y_W} (1 - \tau_W)^{(N-x_E)-y_W} \cdot \delta(1 - \delta)^9 \end{aligned}$$

6. Having found the full conditional distribution in question 5, we can now construct conditional distributions for p_E and δ .

$$\begin{aligned} \pi(p_E | x_E, y_E, y_W) &\propto \tau_E^{y_E} (1 - \tau_E)^{x_E-y_E} \cdot \tau_W^{y_W} (1 - \tau_W)^{(N-x_E)-y_W} \cdot \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \\ &\propto \left(\frac{(1-\delta)p_E}{\theta_E} \right)^{y_E} \left(1 - \frac{(1-\delta)p_E}{\theta_E} \right)^{x_E-y_E} \cdot \left(\frac{(1-\delta)(1-p_E)}{(1-\theta_E)} \right)^{y_W} \left(\frac{\theta_W - (1-\delta)(1-p_E)}{\theta_W} \right)^{(N-x_E)-y_W} \cdot \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \\ &\propto \theta_E^{-y_E} [(1-\delta)p_E]^{y_E} \cdot \theta_E^{-x_E+y_E} [\theta_E - (1-\delta)p_E]^{x_E-y_E} \cdot (1-\theta_E)^{-y_W} (1-\delta)^{y_W} (1-p_E)^{y_W} (1-\theta_E)^{-N+x_E+y_W} \\ &\quad \cdot [\theta_W - (1-\delta)(1-p_E)]^{(N-x_E)-y_W} \cdot \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \\ &\propto \theta_E^{-x_E} (1-\delta)^{y_E} p_E^{y_E} \cdot (1-p_E)^{x_E-y_E} \delta^{x_E-y_E} \cdot (1-\theta_E)^{-y_W} (1-\delta)^{y_W} (1-p_E)^{y_W} (1-\theta_E)^{-N+x_E+y_W} p_E^{(N-x_E)-y_W} \\ &\quad \cdot \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \\ &\propto \theta_E^{-x_E} p_E^{y_E} (1-p_E)^{x_E-y_E} (1-\theta_E)^{-y_W} (1-p_E)^{y_W} (1-\theta_E)^{-N+x_E+y_W} p_E^{(N-x_E)-y_W} \cdot \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \\ &\propto p_E^{y_E+N-x_E-y_W} (1-p_E)^{x_E-y_E+y_W} (1-\theta_E)^{-N+x_E+y_W-y_W+N-x_E} \\ &\propto p_E^{y_E+N-x_E-y_W} (1-p_E)^{x_E-y_E+y_W} \\ &\sim \text{Beta}(N - x_E + y_E - y_W + 1, x_E - y_E + y_W + 1) \end{aligned}$$

$$\begin{aligned} \pi(\delta | x_E, y_E, y_W) &\propto \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \cdot \tau_E^{y_E} (1 - \tau_E)^{x_E-y_E} \cdot \tau_W^{y_W} (1 - \tau_W)^{(N-x_E)-y_W} \cdot \delta(1 - \delta)^9 \\ &\propto \left(\frac{(1-\delta)p_E}{\theta_E} \right)^{y_E} \left(1 - \frac{(1-\delta)p_E}{\theta_E} \right)^{x_E-y_E} \cdot \left(\frac{(1-\delta)(1-p_E)}{(1-\theta_E)} \right)^{y_W} \left(\frac{\theta_W - (1-\delta)(1-p_E)}{\theta_W} \right)^{(N-x_E)-y_W} \cdot \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \cdot \delta(1 - \delta)^9 \\ &\propto \theta_E^{-y_E} [(1-\delta)p_E]^{y_E} \cdot \theta_E^{-x_E+y_E} [\theta_E - (1-\delta)p_E]^{x_E-y_E} \cdot (1-\theta_E)^{-y_W} (1-\delta)^{y_W} (1-p_E)^{y_W} (1-\theta_E)^{-N+x_E+y_W} \\ &\quad \cdot [(1-\delta)(1-p_E) + p_E \delta - (1-\delta)(1-p_E)]^{(N-x_E)-y_W} \cdot \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \cdot \delta(1 - \delta)^9 \text{ (similar to the previous part)} \\ &\propto \theta_E^{-y_E-x_E+y_E+x_E} (1-\delta)^{y_E} p_E^{y_E} (1-p_E)^{x_E-y_E} \delta^{x_E-y_E} (1-\theta_E)^{-N+x_E+N-x_E} (1-\delta)^{y_W} (1-p_E)^{y_W} p_E^{N-x_E-y_W} \cdot \delta(1 - \delta)^9 \\ &\propto \delta^{x_E-y_E+N-x_E-y_W+1} (1-\delta)^{y_E+y_W+9} \\ &\sim \text{Beta}(N - y_W - y_E + 2, y_E + y_W + 10) \end{aligned}$$

7. In order to perform the Gibbs sampling algorithm, we first need to obtain $\pi(p_E, \delta | x_E)$:

$$\begin{aligned}\pi(p_E, \delta | x_E) &= L(X_E; p_E, \delta) \cdot \pi(\delta) \cdot \pi(p_E) \\ &= \binom{N}{x_E} \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \cdot \frac{\Gamma(12)}{\Gamma(2)\Gamma(10)} \delta(1 - \delta)^9 \cdot 1 \\ &\propto \theta_E^{x_E} (1 - \theta_E)^{N-x_E} \cdot \delta(1 - \delta)^9 \Rightarrow \text{non-standard distribution}\end{aligned}$$

Knowing that $Y_E | (X_E, p_E, \delta) \sim \text{Bin}(X_E, \tau_E)$ and $Y_W | (X_E, p_E, \delta) \sim \text{Bin}(N - X_E, \tau_W)$, we can perform the Gibbs sampling algorithm to investigate $\pi(p_E, \delta | x_E)$ with data augmentation using unobserved data y_E and y_W . Sampling Y_E, Y_W, p_E, δ , we obtain a Markov chain which converges to $\pi(Y_E, Y_W, p_E, \delta | x_E)$ – a joint posterior distribution. After a certain number of iterations, the marginal distribution of (p_E, δ) converges to the posterior $\pi(p_E, \delta | x_E)$.

Gibbs algorithm with data augmentation:

1. Initialization
 - 1) Generate the initial value for p_E from its prior distribution $p_E \sim U(0, 1)$.
 - 2) Generate the initial value for δ from its prior distribution $\delta \sim \text{Beta}(2, 10)$.
2. Iterate N times, for $n = 0, 1, \dots, N$
 - 1) Calculate: $\theta_E^{(n)} = p_E^{(n)} * (1 - \delta^{(n)}) * (1 - p_E^{(n)}) * \delta^{(n)}$
 - 2) Calculate: $\tau_E^{(n)} = \frac{(1 - \delta^{(n)}) * p_E^{(n)}}{\theta_E^{(n)}}$
 - 3) Calculate: $\tau_W^{(n)} = \frac{(1 - \delta^{(n)}) * (1 - p_E^{(n)})}{1 - \theta_E^{(n)}}$
 - 4) Simulate the variable $Y_E^{(n)}$ using data augmentation from the $\text{Bin}(X_E, \tau_E^{(n)})$
 - 5) Simulate the variable $Y_W^{(n)}$ using data augmentation from the $\text{Bin}(N - X_E, \tau_W^{(n)})$
 - 6) Sample $p_E^{(n+1)}$ from $\text{Beta}(N - x_E - y_W^{(n)} + y_E^{(n)} + 1, x_E - y_E^{(n)} + y_W^{(n)} + 1)$, given $Y_E^{(n)}$ and $Y_W^{(n)}$
 - 7) Sample $\delta^{(n+1)}$ from $\text{Beta}(N - y_E^{(n)} - y_W^{(n)} + 2, y_E^{(n)} + y_W^{(n)} + 10)$, given $Y_E^{(n)}$ and $Y_W^{(n)}$.

8. The initial values for the Gibbs sampling for the variables p_E and δ are obtained from their prior distributions:

$$\begin{aligned}\pi(p_E) &= p_E \sim U(0, 1) \\ \pi(\delta) &= \delta \sim \text{Beta}(2, 10)\end{aligned}$$

Summary of p_E :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000015	0.287293	0.351223	0.318162	0.383871	0.469199

Summary of δ :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001422	0.119173	0.195979	0.205502	0.286599	0.535632

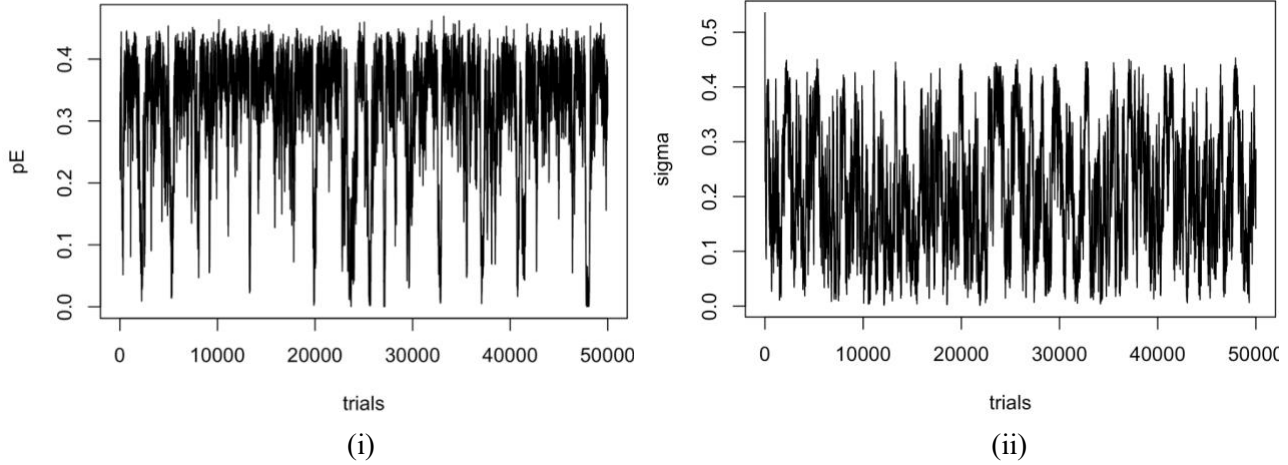


Figure 1: (i) Trace plot of p_E simulated from the conditional distribution; (ii) Trace plot of δ simulated from the conditional distribution

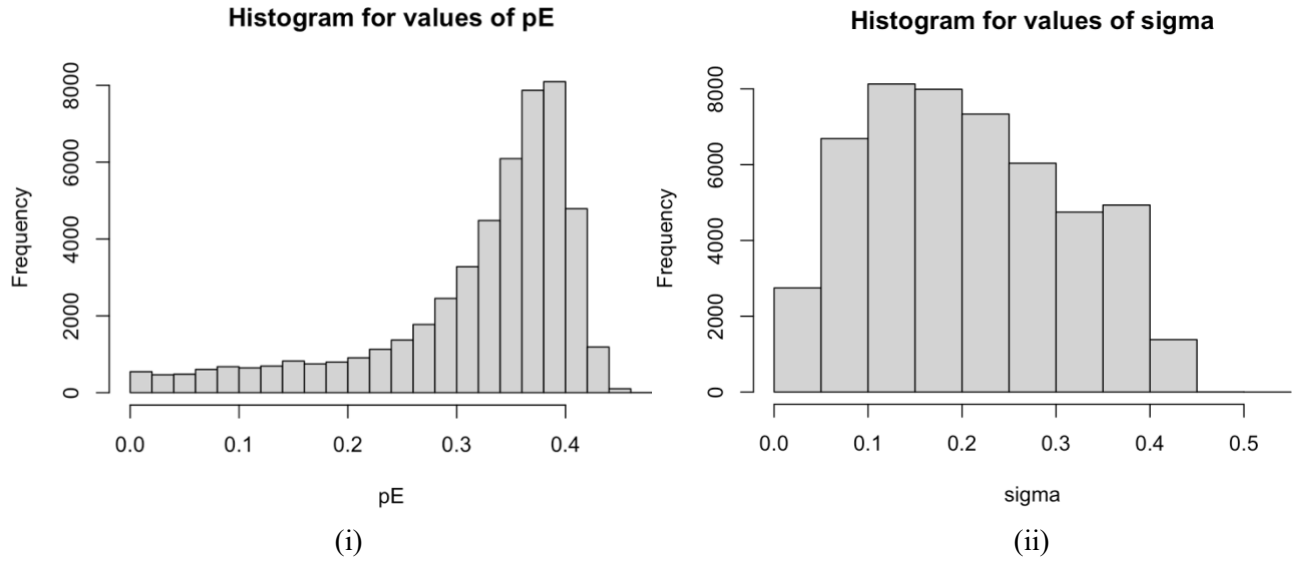


Figure 2: (i) Histogram of p_E simulated from the conditional distribution; (ii) Histogram of δ simulated from the conditional distribution

Comment for p_E : The proportion of seabirds nesting on island E in year 1 (p_E) approximately follows Beta distribution. According to Figure 2(i), the histogram of p_E is skewed to the left. Based on Figure 1(i), the trace plot of p_E has high variability and mostly considers the lower values while being centered around the mean (from the summary the mean = 0.32), therefore the Markov chain appears to explore a range of values.

Comment on δ : The probability that a seabird will change from island one island to another (δ) follows a Beta distribution. According to Figure 2(ii), the histogram of δ appears to be slightly skewed to the right. Based on the summary above, δ has a mean of 0.21, therefore Figure 1(ii) presents the trace plot of δ which also displays high variability and captures the values from the same distance around the mean. This indicates that the posterior distribution of δ has a better understanding of the variable compared to the posterior distribution of p_E .

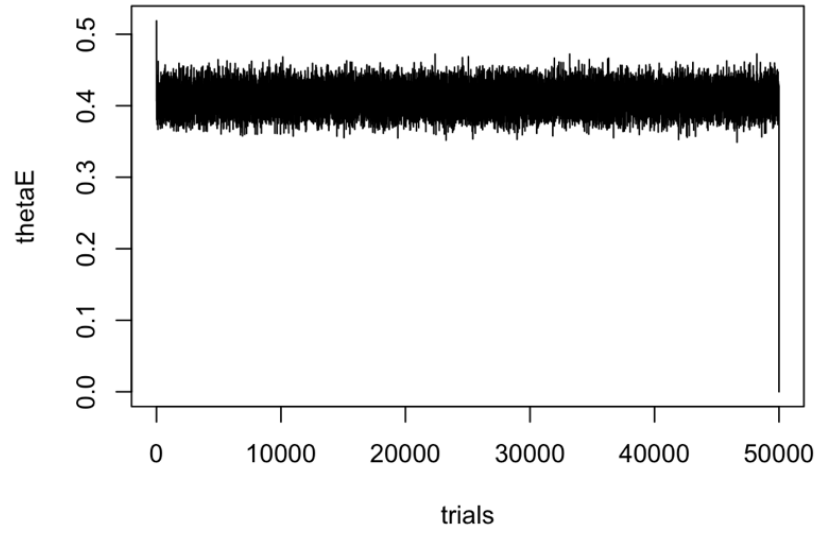


Figure 3: Traceplot of θ_E

Figure 3 is representing the values of θ_E that were used in the simulation process. According to the graph, θ_E mostly takes the values in the range from 3.8 to 4.2. Considering this information, along with the means of p_E and δ observed from the Gibbs sampler, it can be concluded that most of the seabirds that are on island E in year 2, have been there since year 1.